PADHAKU
Science · Class 10 Science



CHAPTER 13

Our Environment

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.
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Activity 13.1 · 13.1 Eco-system — What are its Components?

Activity 13.1 (a) **If we add a few aquatic plants and animals to the aquarium, it can become a self-sustaining system. Can you think how this happens?**

✔ ANSWER

- Aquatic plants make food by photosynthesis and give out oxygen; fish use this oxygen and some plants as food.
- Fish give out carbon dioxide and wastes; plants use the CO_2 and the minerals from broken-down waste.
- Small microorganisms (decomposers) break down wastes and dead matter.

So producers, consumers and decomposers keep supporting each other, and the system runs on its own like a natural **ecosystem** (human-made).

Activity 13.1 (b) **Can we leave the aquarium as such after we set it up? Why does it have to be cleaned once in a while? Do we have to clean ponds or lakes in the same manner? Why or why not?**

✔ ANSWER

- **No.** An aquarium is a small, closed, artificial ecosystem with few decomposers. Uneaten food, fish waste and dead matter collect faster than they are broken down, so the water becomes dirty and harmful. So it must be cleaned now and then.
- **Ponds and lakes need not be cleaned in this way.** They are natural ecosystems with many decomposers (bacteria, fungi) and a complete balance of producers, consumers and decomposers, so wastes are broken down and nutrients recycled naturally.

? In-text questions · Producers, consumers and decomposers

? Consumers can be classed as herbivores, carnivores, omnivores and parasites. Give examples for each of these categories.

✔ ANSWER

- **Herbivores:** cow, goat, deer, rabbit
- **Carnivores:** lion, tiger, frog, snake
- **Omnivores:** human, crow, bear, cockroach
- **Parasites:** louse, tapeworm, Cuscuta (amarbel), mosquito

? What will happen to the garbage, and dead animals and plants in the absence of decomposers? Will the natural replenishment

of the soil take place, even if decomposers are not there?

✔ ANSWER

- Garbage, dead plants and animals would pile up and not break down; the surroundings would become dirty and smelly.
- **No.** Without decomposers, the complex organic substances are not broken into simple inorganic substances, so nutrients would not return to the soil. The soil would not be replenished and plants would not get minerals.

🧪 Activity 13.2

Activity 13.2 (a) **While creating an aquarium did you take care not to put an aquatic animal which would eat others? What would have happened otherwise?**

✔ ANSWER

Yes. If a predator (carnivorous) fish was put in, it would eat the other fish. Soon the other fish would be finished, and finally the predator would also die for lack of food. The balance of the aquarium would break down.

Activity 13.2 (b) **Make groups and discuss how each of the above groups of organisms**

are dependent on each other.

💡 SAMPLE ANSWER

- **Producers** (plants) make food and oxygen; all consumers depend on them directly or indirectly.
- **Consumers** depend on producers or other consumers for food; they also give CO_2 used by plants.
- **Decomposers** depend on dead remains and wastes of producers and consumers; they return nutrients to the soil/water for producers.

So all three groups depend on one another.

Activity 13.2 (c) **Write the aquatic organisms in order of who eats whom and form a chain of at least three steps.**

✔ ANSWER

Phytoplankton (algae) → Zooplankton → Small fish → Large fish

Activity 13.2 (d) **Would you consider any one group of organisms to be of primary importance? Why or why not?**

💡 SAMPLE ANSWER

No single group can be called the only important one: all groups depend on each other. Producers are the source of food and energy, consumers keep populations in check, and decomposers recycle nutrients. If any one group is removed, the ecosystem is disturbed.

(Producers may be called the base, since all energy enters the ecosystem through them, but they also need decomposers to get nutrients back.)

🧪 Activity 13.3 · 13.1.1 Food Chains and Webs

Activity 13.3 (a) **Newspaper reports about pesticide levels in ready-made food items are often seen and some states have banned these products. Debate in groups the need for such bans.**

💡 SAMPLE ANSWER

Sample points: such bans are needed because pesticides are non-biodegradable, get accumulated in our bodies by biological magnification and can cause diseases. Bans force companies to use safe raw materials. Against: bans affect business, so testing and fixed safe limits may be a better long-term step.

Activity 13.3 (b) **What do you think would be the source of pesticides in these food**

items? Could pesticides get into our bodies from this source through other food products too?

✔ ANSWER

- Source: pesticides sprayed on crops (grains, fruits, vegetables) used as raw material; they reach soil and water and are absorbed by plants.
- **Yes.** They can also enter our bodies through other foods: grains, vegetables, fruits, milk, meat and fish, because they move up the food chain.

Activity 13.3 (c)

Discuss what methods could be applied to reduce our intake of pesticides.

💡 SAMPLE ANSWER

- Wash fruits and vegetables well; peel where possible.
- Use organic farming and biopesticides/natural pest control.
- Use only permitted pesticides in the right amounts.
- Regular testing of food items and banning unsafe products.

Questions · 13.1 Eco-system — What are its Components?

Q 1 **What are trophic levels? Give an example of a food chain and state the different trophic levels in it.**

✔ ANSWER

Each step or level of a food chain, at which energy is transferred, is called a **trophic level**.

Example: **Grass → Grasshopper → Frog → Snake → Hawk**

- Grass: first trophic level (producer)
- Grasshopper: second trophic level (herbivore / primary consumer)
- Frog: third trophic level (small carnivore / secondary consumer)
- Snake: fourth trophic level (larger carnivore / tertiary consumer)
- Hawk: fifth trophic level (top carnivore)

Q 2 **What is the role of decomposers in the ecosystem?**

✔ ANSWER

- Decomposers (bacteria and fungi) break down dead remains and waste products of organisms.
- They change complex organic substances into simple inorganic substances that go into the soil and are used again by plants. So they **recycle nutrients** and keep the soil fertile.

- They also clean the environment by removing dead bodies and wastes.

Activity 13.4 · 13.2.1 Ozone Layer and How it is Getting Depleted

Activity 13.4 (a) **Find out which chemicals are responsible for the depletion of the ozone layer.**

✔ ANSWER

Mainly **chlorofluorocarbons (CFCs)**, used in refrigerators, air conditioners and aerosol sprays; also halons (used in fire extinguishers), carbon tetrachloride, methyl chloroform and methyl bromide.

Activity 13.4 (b) **Find out if the regulations put in place to control the emission of these chemicals have succeeded in reducing the damage to the ozone layer. Has the size of the hole in the ozone layer changed in recent years?**

💡 SAMPLE ANSWER

Yes, they have largely succeeded. After the 1987 agreement (Montreal Protocol, UNEP) to freeze and then phase out CFCs, their amount in the atmosphere has been falling. Scientists report that the ozone hole over Antarctica is slowly getting smaller and the layer is expected to recover in the coming decades, though its size still changes from year to year.

Activity 13.5 · 13.2.2 Managing the Garbage we Produce

Activity 13.5 (a) **Bury household waste in a pit (cover with 15 cm of soil), keep it moist and observe at 15-day intervals. What are the materials that remain unchanged over long periods of time?**

✔ ANSWER

Plastic (milk packets, polythene), glass/empty medicine bottles, medicine strips and bubble packs, rubber/plastic footwear and synthetic cloth remain unchanged.

Activity 13.5 (b) **What are the materials which change their form and structure over time?**

✔ ANSWER

Kitchen waste (spoilt food, vegetable peels, used tea leaves), paper and cartons, cotton cloth and leather slowly change their form and rot.

Activity 13.5 (c) **Of these materials that are changed, which ones change the fastest?**

✔ ANSWER

Kitchen waste: spoilt food, vegetable peels and used tea leaves change the fastest.

Conclusion: substances broken down by microorganisms are **biodegradable**; others are **non-biodegradable**.

? **How many of the substances you buried were biodegradable?**

💡 SAMPLE ANSWER

Kitchen waste, paper, cartons, cotton cloth and leather were biodegradable; plastics, glass, medicine strips and synthetic items were not. (Count from your own pit.)

🧪 Activity 13.6

Activity 13.6 (a) **Use the library or internet to find out more about biodegradable and non-biodegradable substances.**

✔ ANSWER

- **Biodegradable:** broken down by bacteria and other saprophytes, e.g. food, paper, wood, cotton, dung.
- **Non-biodegradable:** not broken down by biological processes, e.g. plastics, glass, metals, DDT, synthetic fibres.

Activity 13.6 (b) **How long are various non-biodegradable substances expected to last in our environment?**

💡 SAMPLE ANSWER

Approximate times: plastic bags about 20–1000 years, plastic bottles about 450 years, aluminium cans about 200 years, glass bottles about a million years (practically forever). They persist for a very long time.

Activity 13.6 (c) **New types of plastics said to be biodegradable are available. Find out more about such materials and whether they do or do not harm the environment.**

💡 SAMPLE ANSWER

Biodegradable plastics are made from plant materials like corn starch or sugarcane (e.g. PLA). They break down faster, but many need high temperature and special composting conditions. In open soil or water they may break into tiny pieces (microplastics) that still harm living things. So they are better than ordinary plastic but not fully harmless.

? Questions · 13.2 How do our Activities Affect the Environment? (1)

Q 1 **Why are some substances biodegradable and some non-biodegradable?**

✔ ANSWER

Enzymes of bacteria and other saprophytes are specific in their action; they can break down only certain substances.

- Substances (like food, paper, wood) that these microorganisms can break down are **biodegradable**.

- Substances (like plastics, glass, DDT) that they cannot break down are **non-biodegradable**; they persist for a long time.

Q 2 **Give any two ways in which biodegradable substances would affect the environment.**

✔ ANSWER

- When broken down, they return nutrients to the soil and increase its fertility (manure/compost).
- If they collect in large amounts, they rot and give a foul smell, attract flies and germs and spread diseases.

Q 3 **Give any two ways in which non-biodegradable substances would affect the environment.**

✔ ANSWER

- They persist for a long time and pollute soil and water; they block drains and reduce soil fertility.
- Some enter the food chain and accumulate at higher levels (biological magnification), harming organisms; animals may also die by eating plastic.

Activity 13.7

Activity 13.7 (a) **Find out what happens to the waste generated at home. Is there a system in place to collect this waste?**

💡 SAMPLE ANSWER

Sample: waste is collected by the municipal / panchayat garbage van or a sweeper daily and taken to a dump site (landfill). Some homes keep separate dustbins for wet and dry waste.

Activity 13.7 (b) **Find out how the local body deals with the waste. Are there mechanisms to treat biodegradable and non-biodegradable wastes separately?**

💡 SAMPLE ANSWER

Sample: the municipal body asks people to separate wet (biodegradable) and dry (non-biodegradable) waste. Wet waste goes for composting; dry waste is sorted for recycling and the rest goes to landfill. In many areas this separation is still not done fully.

Activity 13.7 (c) **Calculate how much waste is generated at home in a day. How much of this is biodegradable?**

💡 SAMPLE ANSWER

Sample: a family of 5 may make about 1.5–2 kg of waste a day, of which about 60–70% (kitchen waste, paper) is biodegradable. (Weigh your own home's waste.)

Activity 13.7 (d) **Calculate how much waste is generated in the classroom in a day. How much of this is biodegradable?**

💡 SAMPLE ANSWER

Sample: about 0.5 kg a day, mostly paper and food leftovers (biodegradable), with some wrappers, pens and plastic bottles (non-biodegradable). (Weigh your own class's waste.)

Activity 13.7 (e) **Suggest ways of dealing with this waste.**

💡 SAMPLE ANSWER

- Separate wet and dry waste in different dustbins.
- Make compost from kitchen waste and leaves.
- Recycle paper, plastic, glass and metal; reuse things where possible.

- Reduce use of plastic and disposable items.

🧪 Activity 13.8

Activity 13.8 (a) **Find out how the sewage in your locality is treated. Are there mechanisms to ensure that local water bodies are not polluted by untreated sewage?**

💡 SAMPLE ANSWER

Sample: sewage flows through underground drains to a sewage treatment plant, where solids are removed and the water is treated before release into a river. In some areas untreated sewage still enters drains and water bodies, so more treatment plants are needed.

Activity 13.8 (b) **Find out how the local industries treat their wastes. Are there mechanisms to ensure that soil and water are not polluted?**

💡 SAMPLE ANSWER

Sample: industries are required by the Pollution Control Board to treat their effluents in effluent treatment plants before release. Some do this properly; others release untreated waste, so regular checking is needed.

❓ In-text questions · Life-style and waste

? **Improvements in life-style, more disposable things and changes in packaging have made much of our waste non-biodegradable. What do you think will be the impact of these on our environment?**

💡 **SAMPLE ANSWER**

More waste will be produced and most of it will not break down. It will pile up in landfills, pollute soil and water, block drains, harm animals that eat it, and some harmful chemicals may enter food chains. Managing waste will become harder and costlier.

Think it over **Disposable cups in trains: what do you think are the advantages of disposable paper-cups over disposable plastic cups?**

✔ **ANSWER**

- Paper cups are **biodegradable**; they break down quickly, while plastic cups persist for a long time.
- Paper can be recycled and does not release harmful chemicals into soil like plastic.
- Unlike kulhads, making them does not remove fertile top-soil.
- They are hygienic as they are used only once.

Activity 13.9

Activity 13.9 (a) **Find out what hazardous materials have to be dealt with while disposing of electronic items. How would these materials affect the environment?**

💡 **SAMPLE ANSWER**

E-waste contains lead, mercury, cadmium, chromium, arsenic, brominated flame retardants and PVC plastic. If dumped or burnt, these poison soil and groundwater, pollute air and can enter food chains, causing damage to the brain, kidneys and nerves of humans and animals.

Activity 13.9 (b) **Find out how plastics are recycled. Does the recycling process have any impact on the environment?**

💡 **SAMPLE ANSWER**

Plastics are collected, sorted by type, washed, shredded, melted and made into pellets used to make new items.

Yes, there is some impact: recycling uses energy and water, and melting may release harmful gases if not done properly. Each time, plastic quality falls, so it cannot be recycled forever. Still, it is better than dumping or burning.

Questions · 13.2 How do our Activities Affect the Environment? (2)

Q 1 **What is ozone and how does it affect any ecosystem?**

✔ ANSWER

Ozone (O_3) is a molecule made of three atoms of oxygen. At ground level it is a deadly poison.

At higher levels of the atmosphere, the ozone layer shields the earth's surface from harmful **ultraviolet (UV) radiation** of the Sun.

UV radiation causes skin cancer and damages organisms. So ozone protects all living things; if the ozone layer is depleted, more UV rays reach the earth and harm plants, animals and humans, disturbing the ecosystem.

Q 2 **How can you help in reducing the problem of waste disposal? Give any two methods.**

✔ ANSWER

- **Separate** biodegradable and non-biodegradable waste in different dustbins; make compost from kitchen waste.
- **Reduce, reuse and recycle:** carry cloth bags instead of plastic bags, avoid disposable items, and send paper, plastic, glass and metal for recycling.

Exercises

1 **Which of the following groups contain only biodegradable items?**

(a) Grass, flowers and leather

(b) Grass, wood and plastic

(c) Fruit-peels, cake and lime-juice

(d) Cake, wood and grass

✔ ANSWER

(a), (c) and (d). Grass, flowers, leather, fruit-peels, cake, lime-juice and wood are all broken down by microorganisms (leather only slowly). (b) is wrong because it contains plastic, which is non-biodegradable.

2 **Which of the following constitute a food-chain?**

(a) Grass, wheat and mango

(b) Grass, goat and human

(c) Goat, cow and elephant

(d) Grass, fish and goat

✔ ANSWER

(b) Grass, goat and human: grass (producer) → goat (herbivore) → human (consumer). The others do not show who eats whom.

3 **Which of the following are environment-friendly practices?**

(a) Carrying cloth-bags to put purchases in while shopping

(b) Switching off unnecessary lights and fans

(c) Walking to school instead of getting your mother to drop you on her scooter

(d) All of the above

✔ ANSWER

(d) All of the above: each reduces waste, saves electricity or cuts fuel use and pollution.

4 **What will happen if we kill all the organisms in one trophic level?**

✔ ANSWER

The food chain will break and the ecosystem will be disturbed:

- Organisms of the next higher level will not get food and will die or move away.
- Organisms of the level below will increase greatly, as nothing eats them.

Example: if all herbivores (deer) are killed, plants grow unchecked, and carnivores (tiger) starve. So the balance of nature is lost.

5 **Will the impact of removing all the organisms in a trophic level be different for different trophic levels? Can the organisms of any trophic level be removed without causing any damage to the ecosystem?**

✔ ANSWER

Yes, the impact differs:

- Removing **producers** destroys the whole ecosystem, as no level would get food or energy.
- Removing **herbivores** makes carnivores starve and plants grow unchecked.
- Removing **carnivores** makes herbivores increase too much; they overgraze and destroy plants.
- Removing **decomposers** stops recycling of nutrients; dead matter piles up.

No, organisms of no trophic level can be removed without damaging the ecosystem, because all levels are interdependent.

6 **What is biological magnification? Will the levels of this magnification be different at different levels of the ecosystem?**

✔ ANSWER

Biological magnification: the progressive increase in the concentration of harmful non-biodegradable chemicals (like pesticides, e.g. DDT) at each higher trophic level of a food chain.

Yes, the levels are different. They are lowest at the producer level and increase at each step, so they are **highest at the top** of the food chain. Humans, at the top, get the maximum concentration.

7 **What are the problems caused by the non-biodegradable wastes that we generate?**

✔ ANSWER

- They persist for a long time and pile up, making the surroundings dirty.
- They pollute soil and water and reduce soil fertility.
- Plastic blocks drains and sewers, causing waterlogging and breeding of mosquitoes.
- Animals die after eating plastic bags.
- Harmful chemicals (pesticides) enter food chains and get magnified at higher levels.
- Burning them releases poisonous gases into the air.

8 **If all the waste we generate is biodegradable, will this have no impact on the environment?**

✔ ANSWER

No, it will still have an impact, though less harmful:

- Large amounts of waste take time to decompose; while it lies around, it rots and gives a foul smell.
- It attracts flies, mosquitoes and germs, spreading diseases.
- Decomposition releases gases like methane and CO_2 (greenhouse gases).
- Waste in water bodies uses up oxygen and harms aquatic life.

So proper disposal (composting etc.) is still needed.

9 **Why is damage to the ozone layer a cause for concern? What steps are being taken to limit this damage?**

✔ ANSWER

Concern: the ozone layer shields the earth from harmful UV radiation of the Sun. If it is damaged, more UV rays reach the earth, causing skin cancer, eye damage (cataract) and weak immunity in humans, and harming plants and other organisms.

Steps taken:

- In 1987, UNEP made an agreement to freeze CFC production at 1986 levels (and later phase it out).

- It is now mandatory for manufacturing companies all over the world to make **CFC-free refrigerators**.
- Use of CFCs in aerosols and fire extinguishers is being replaced with safer substitutes.

🎉 **Chapter 13 done! Revise the answers once more tomorrow.**